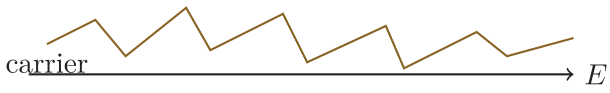
Where Does Ohm's Law Come From?

Name: ____________________

A law usually handed over as voltage equals current times resistance, rebuilt instead from a single charge carrier bouncing through a lattice.



The question. What makes a charge carrier drift, what slows it down, and why does it not simply accelerate forever?

Working rule: $V = IR$ is not the starting point. No claim is made that Ohm's law is a law of nature in the sense Coulomb's is.

1. What makes it drift? — a steady force

$$a = \frac{qE}{m}$$

Left alone, what would this predict for the carrier's speed over time? Why is that plainly not what happens in an ordinary wire?

2. What slows it down? — collisions

Describe, in your own words, what a collision does to a carrier's velocity, and why repeated collisions prevent unbounded acceleration.

3. From acceleration to drift — average over many collisions

Let τ be the average time between collisions. Show that the mean drift velocity gained before the next collision is

$$v_d = a\tau = \underline{\hspace{6cm}}.$$

Define mobility μ so that $v_d = \mu E$.

$$\mu = \underline{\hspace{5cm}}$$

4. From drift to current — current density

Current density is charge per carrier, times carrier density n, times drift velocity.

$$J = nqv_d = \underline{\hspace{6cm}} = \sigma E$$

Identify σ in terms of n, q, τ, and m.

$$\sigma = \underline{\hspace{5cm}}$$

5. Ohm's law as a constitutive law — interaction or constitutive?

List every quantity in your expression for σ. For each, state whether it is a universal constant or a property of the particular material.

Quantity	Universal or material-specific?
n	
q	
τ	
m	

6. The Hooke's law parallel — a linear-response approximation

Complete the comparison.

	Ohm's law	Hooke's law
Linear relation	$V = IR$	________________
Breaks down when		
Constant depends on		

7. Core results — what this enquiry actually earned

The relations established along the way.

$$a = \frac{qE}{m} \qquad v_d = \frac{qE\tau}{m} = \mu E \qquad J = nq\mu E = \sigma E \qquad \sigma = \frac{nq^2\tau}{m}$$

Concepts earned, in order.

- A steady force alone predicts unbounded acceleration; collisions are what actually limit it.
- Drift velocity is a small, steady bias riding on top of constant resetting by collisions.
- Mobility and conductivity bundle carrier properties that are specific to the material, not universal.
- Ohm's law is a consequence of a constitutive assumption about a material's response, exactly like Hooke's law, not a fundamental interaction law like Coulomb's.

Every substantial claim should be accompanied by a reason, derivation, or counterexample.